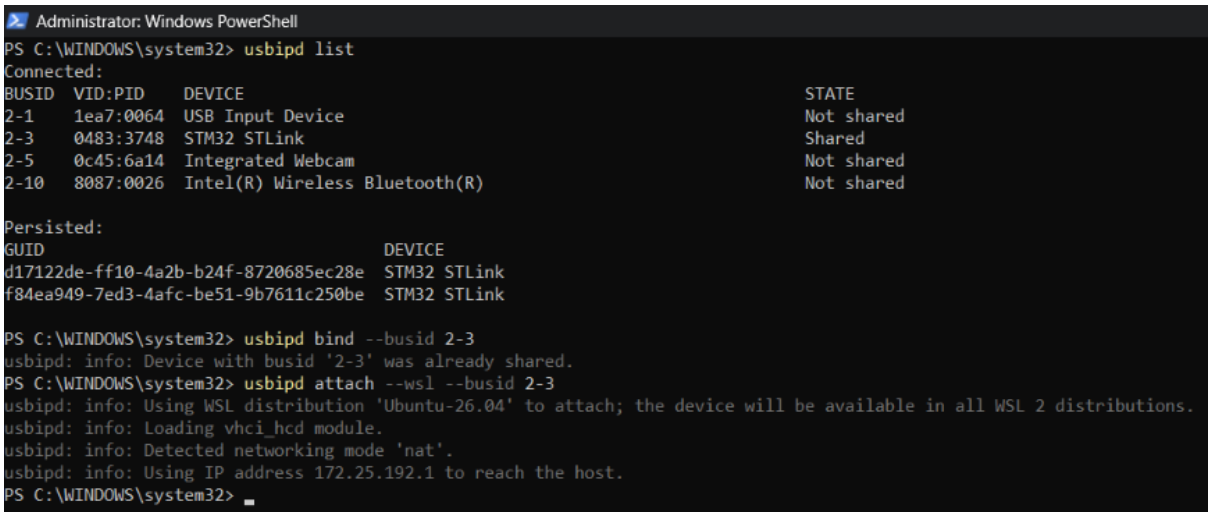
RTOS



Connecting stm32 to wsl and then using it inside it

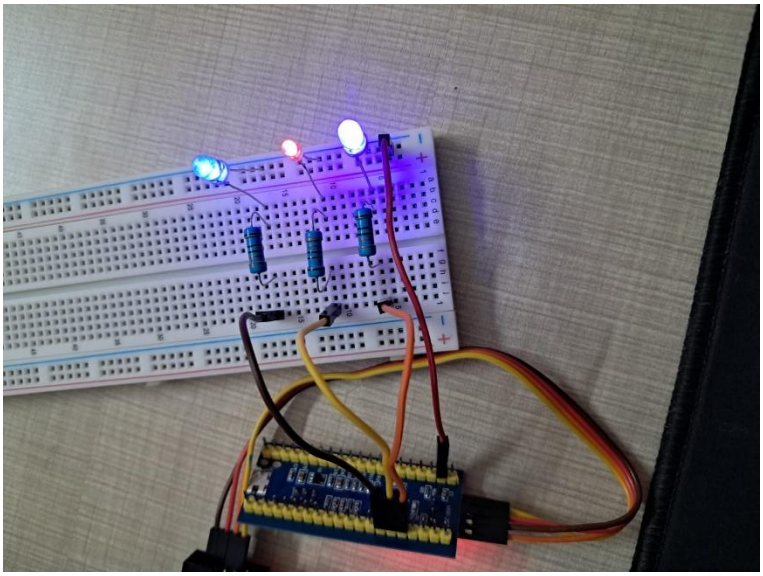
1. Once done with the connection then go to the wsl and see if the device is running using **lsusb** which will show the connected usb devices.
2. Once the device is found there then redirect to the folder where the execution is to be done by you and do the required over there

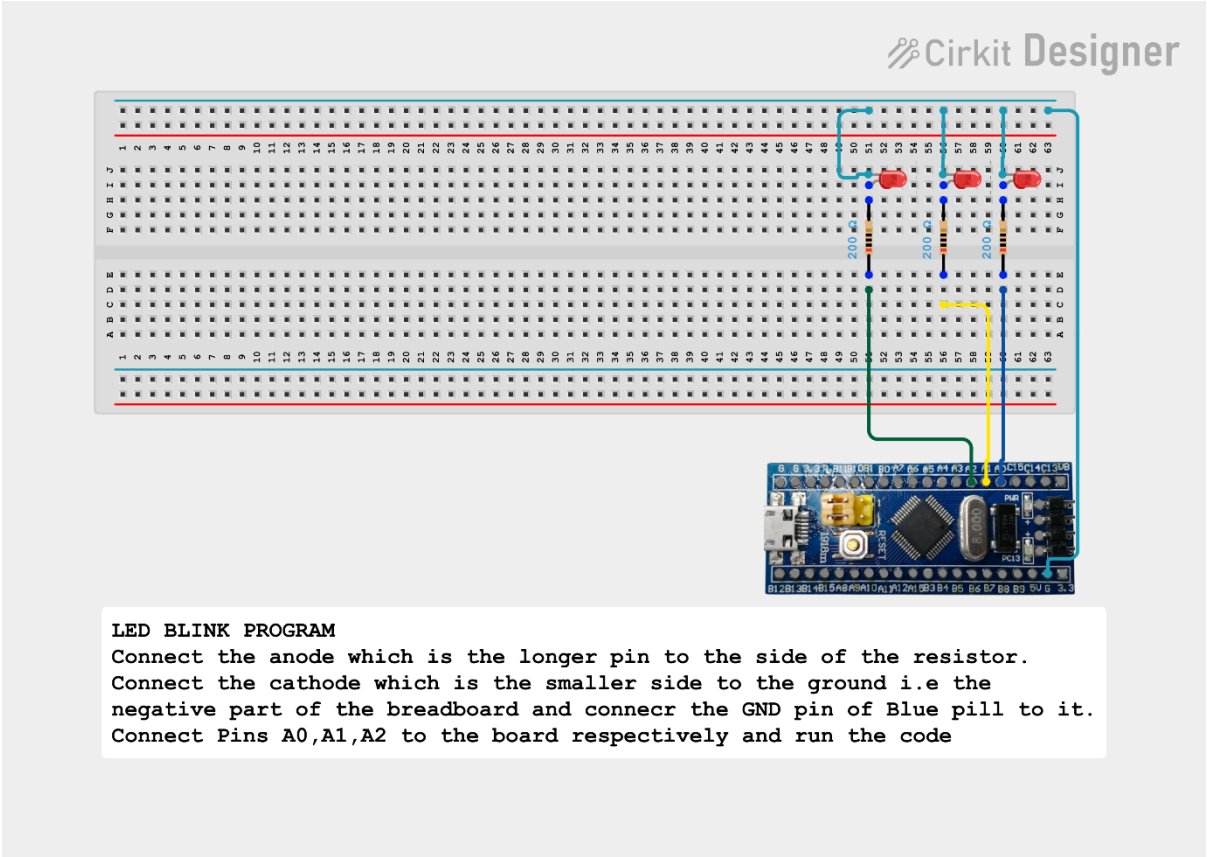
PROGRAM 1 : led – blink

The program runs in a normal way as it is only used to see if PC13 on the board runs well and to check if it glows or not and that works fine and no additional connections are needed for this unless and until if it has to be verified on an external LED.

PROGRAM 2 : Three – LED – BLINK

All the three LED's Blink at once and there is no particular order for them to glow, PC13 is used by default and it can be changed according to the user choice and I have used an external LED

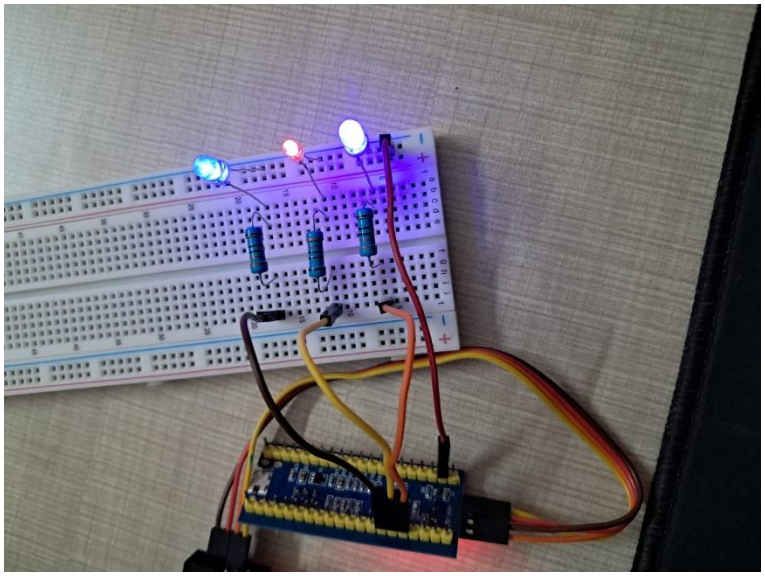


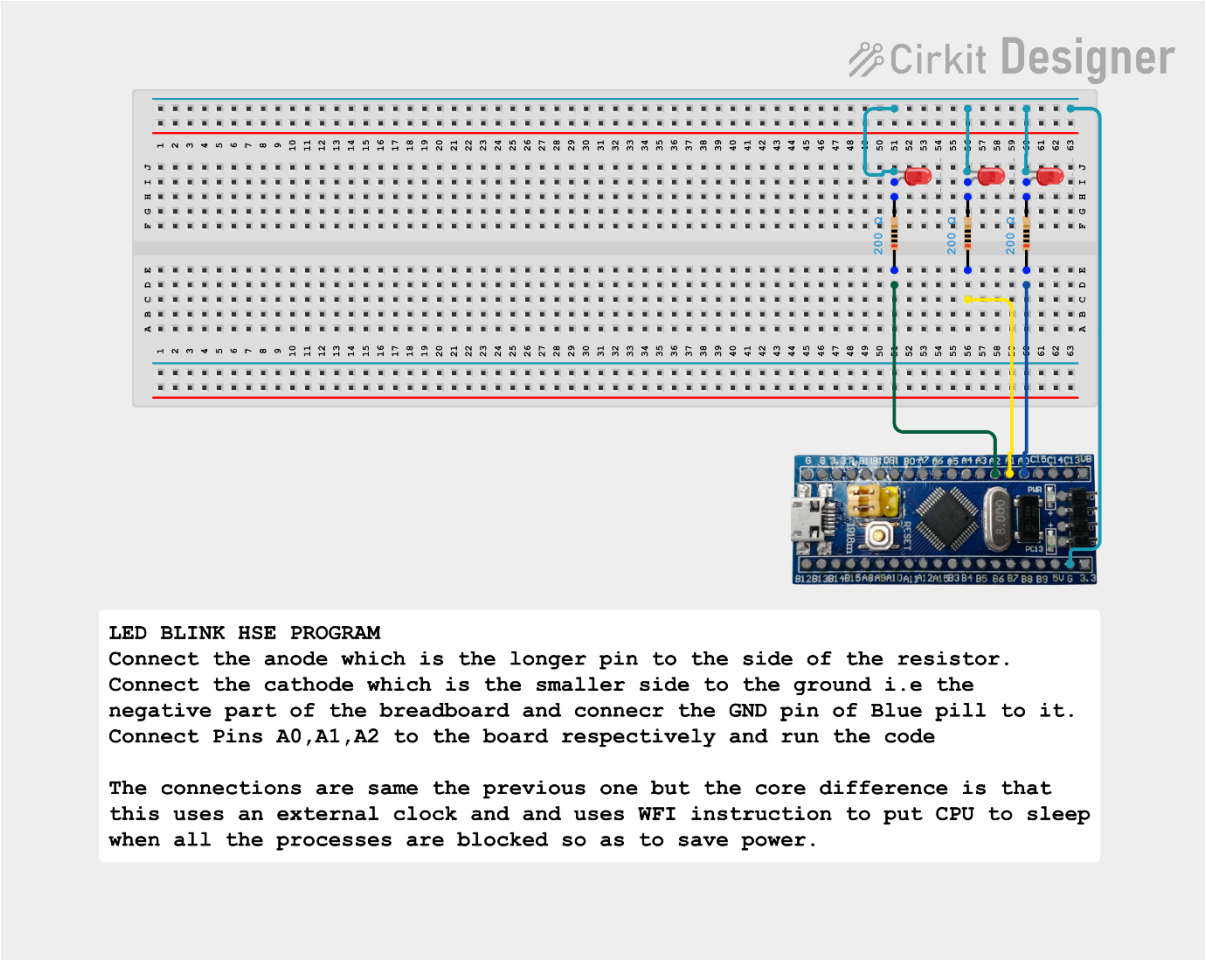



```
LED BLINK PROGRAM
Connect the anode which is the longer pin to the side of the resistor.
Connect the cathode which is the smaller side to the ground i.e the
negative part of the breadboard and connecr the GND pin of Blue pill to it.
Connect Pins A0,A1,A2 to the board respectively and run the code
```

PROGRAM 3 : Three LED BLINK HSE

This is similar to Three LED blink while the only difference is the clock and the WFI which is used to put CPU to sleep when all the processes are blocked so as to save power.



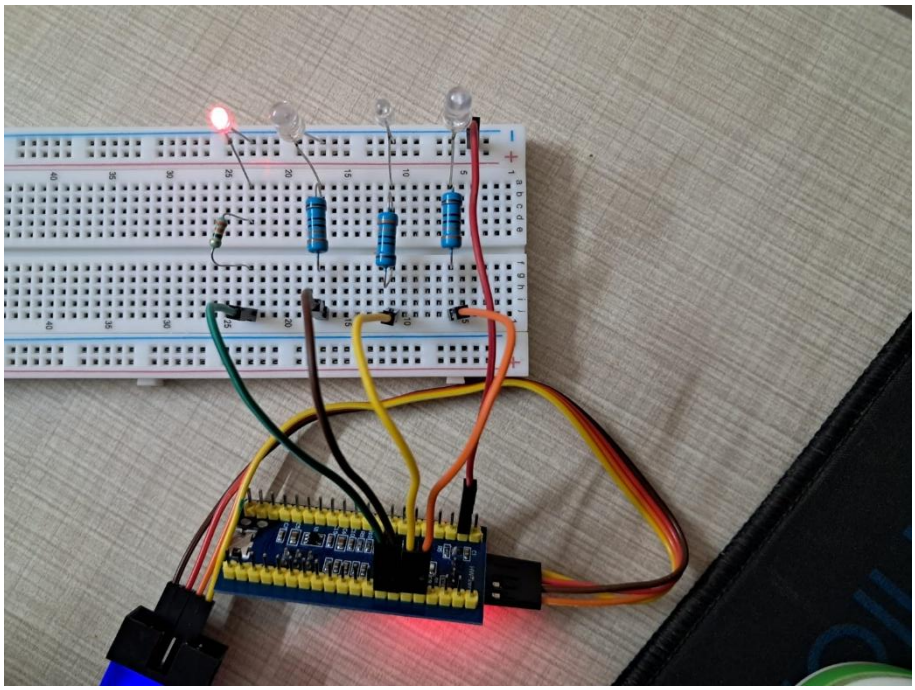


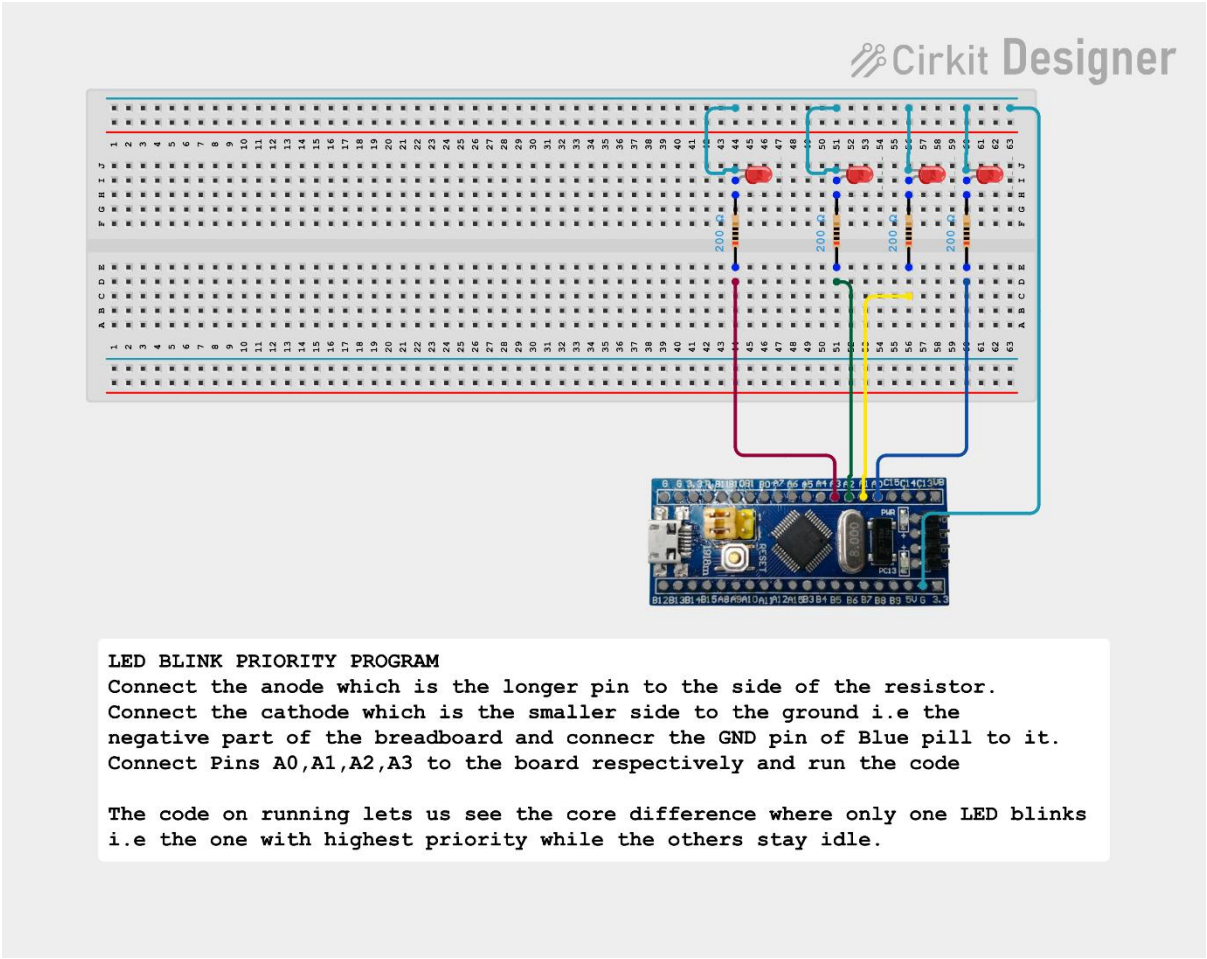

```
LED BLINK HSE PROGRAM
Connect the anode which is the longer pin to the side of the resistor.
Connect the cathode which is the smaller side to the ground i.e the
negative part of the breadboard and connecr the GND pin of Blue pill to it.
Connect Pins A0,A1,A2 to the board respectively and run the code

The connections are same the previous one but the core difference is that
this uses an external clock and and uses WFI instruction to put CPU to sleep
when all the processes are blocked so as to save power.
```

PROGRAM 4 : Three LED Blink Priority

In this one only one LED, ie the led with the highest priority glows while the one with the next priorities don’t glow.



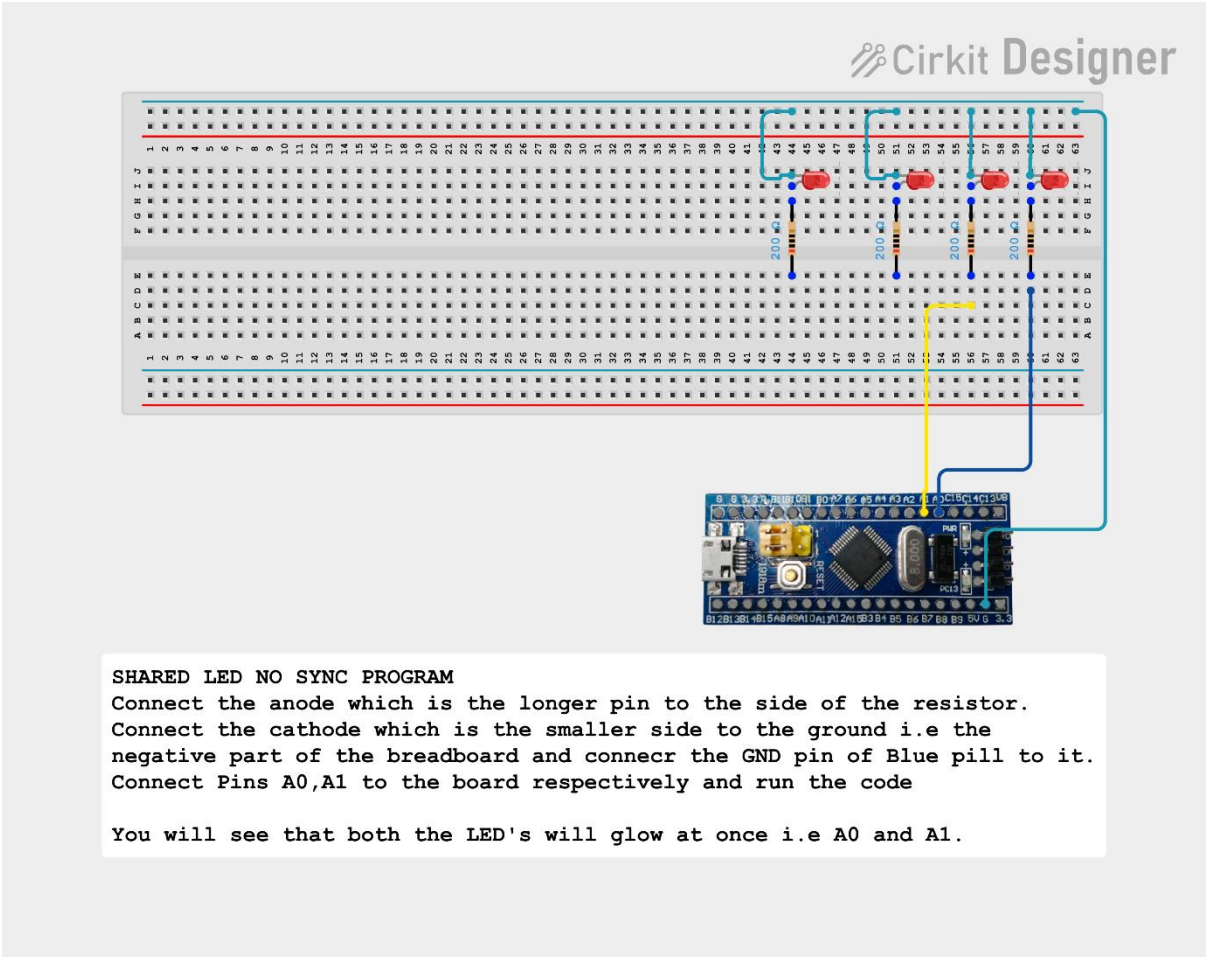



```
LED BLINK PRIORITY PROGRAM
Connect the anode which is the longer pin to the side of the resistor.
Connect the cathode which is the smaller side to the ground i.e the
negative part of the breadboard and connecr the GND pin of Blue pill to it.
Connect Pins A0,A1,A2,A3 to the board respectively and run the code

The code on running lets us see the core difference where only one LED blinks
i.e the one with highest priority while the others stay idle.
```

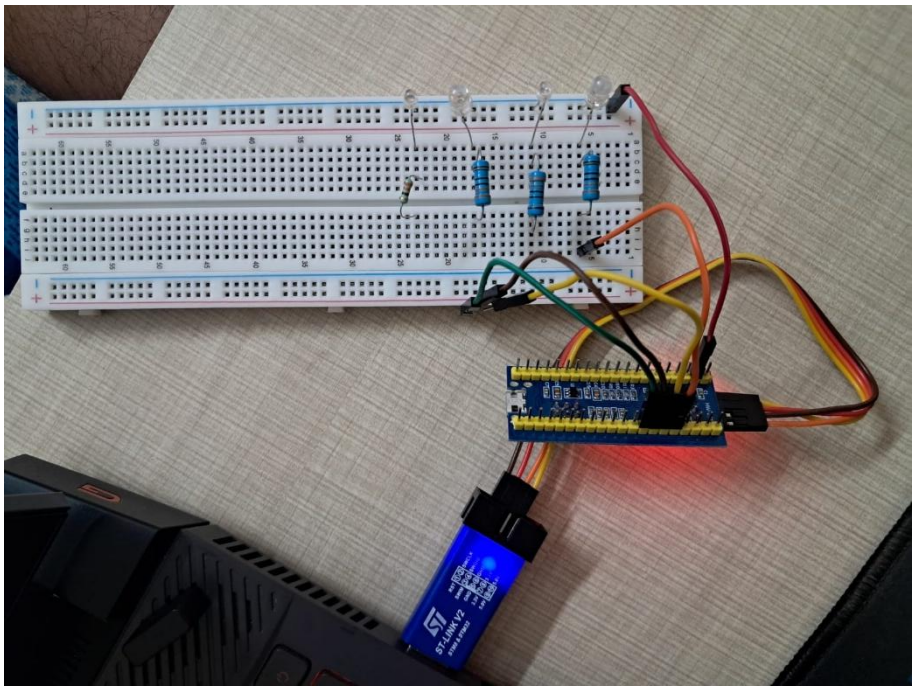
PROGRAM 5 : SHARED LED NO SYNC

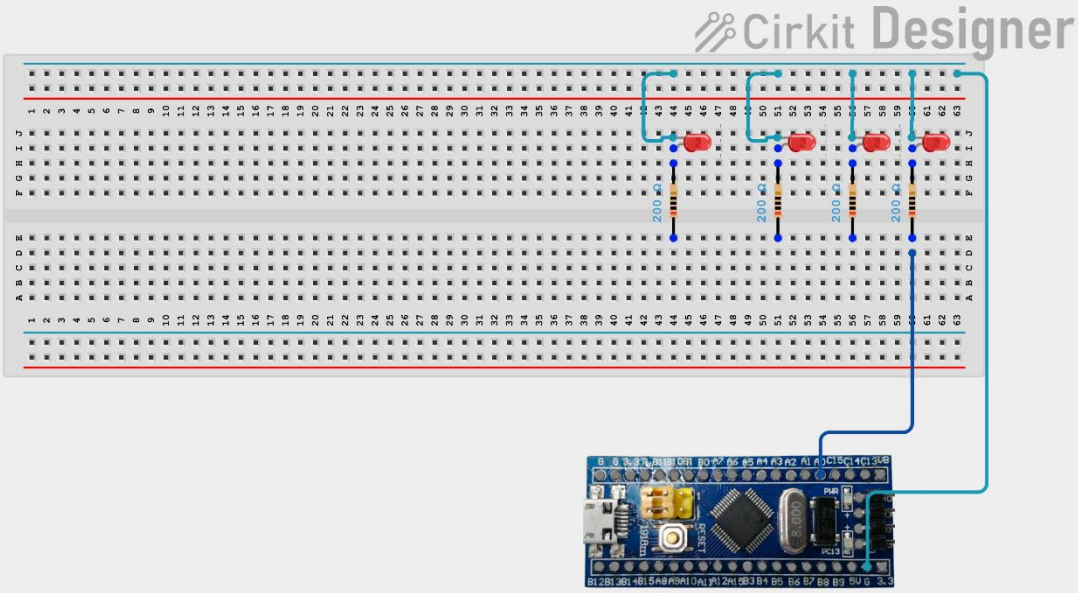
This is similar to the first Three LED blink and here 2 led's blink at once without any synchronization



PROGRAM 6 : SHARED LED MUTEX

In this program two processes try to toggle only one led and due to that the race condition occurs where nothing happens or else they will glow at unwanted intervals or so.





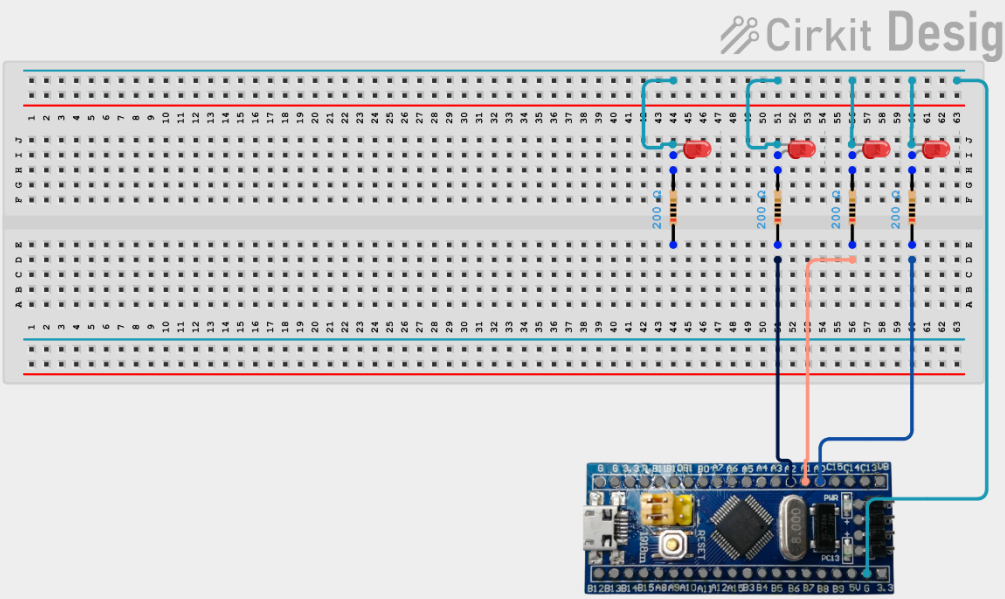
SHARED LED MUTEX PROGRAM
Connect the anode which is the longer pin to the side of the resistor.
Connect the cathode which is the smaller side to the ground i.e the negative part of the breadboard and connecr the GND pin of Blue pill to it.
Connect Pin A0 to the board respectively and run the code

The core idea of this code is to show what happens when the race condition occurs i.e. when both the pins to try to toggle a single one then race condition occurs due tho which the LED might not glow at all or else it might glow at irregular intervals or else at once.

This code is just used to see how race condition occurs and to show the effects of it.

PROGRAM 7 : PRIORITY INVERSION

The code shows how priority inversion is done so as to avoid any race conditions that are about to occur and there are all the 3 leds connected and if seen correctly the led PA1 starts first followed by PA2 and then PA0 goes off and PA0 glows and it is only PA0 and PA2 From that particular time.




```
PRIORITY INVERSION PROGRAM
Connect the anode which is the longer pin to the side of the resistor.
Connect the cathode which is the smaller side to the ground i.e the
negative part of the breadboard and connecr the GND pin of Blue pill to it.
Connect Pin A0,A1,A2 to the board respectively and run the code

The pin A1 is the low priority task and it is given the resource first,
so if the code runs correctly, then you will see that the led blinks for a
while i.e led connected to PA1 and then it goes off and then it stays only
between PA0 and PA2
```